

Cover Letter

To: Editorial Office, *Quantum Science and Technology* (IOP Publishing)
Date: 24 May 2026 **From:** Aleaddin Özer (Corresponding Author),
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Dear Editors,

Please find attached our manuscript, *“Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility,”* which we are submitting to *Quantum Science and Technology*.

The post-quantum migration is now an operational problem rather than a research-only one. NIST has finalised FIPS 203, 204 and 205; the U.S. NSA’s CNSA 2.0 timetable mandates broad adoption between 2030 and 2035; and a parallel hardware track — Quantum Key Distribution, governed by ETSI GS QKD 014 — offers an orthogonal channel-level control on high-assurance links. Yet the tools operators use to inventory their cryptographic posture continue to answer a single question: *“is this asset PQ-ready?”* That framing collapses two assets with very different migration costs into one bucket and is silent on the channel through which their key material travels.

This manuscript treats quantum-risk posture as a three-axis problem — algorithmic resistance (A), channel protection (C), and migration agility (G) — and combines the three with the operator’s deadline horizon into a single deadline-adjusted score $q(\text{asset}, t)$. The contribution is in three layers:

1. A formal three-axis posture model with explicit scoring rubrics, a deadline-tension term that shrinks the agility weight as the migration deadline approaches, and a worked four-asset example.
2. An open observability platform (Sezar) implementing the model end to end — eBPF-based TLS observation, an ETSI GS QKD 014 collector with a reusable Key-Management-Entity emulator, a static crypto-agility scanner, and a unified posture rollup. All under MIT at <https://github.com/e2esolutions-tech/sezar>.
3. Three reproducible empirical studies establishing a first baseline on real corpora — a Tranco-top-1k TLS-handshake survey (43.8% hybrid PQ-KEM adoption on 317 of 724 responsive hosts), a controlled ETSI 014 emulator study characterising 13 induced KME/link failure modes (13/13

classification correctness), and a crypto-agility pilot with 91% agreement against hand-graded ground truth on an eleven-project subset of a fifty-project corpus.

The work sits within *Quantum Science and Technology*'s scope on two counts. The QKD axis treats channel-level quantum-secure key delivery as a first-class observable alongside the algorithmic PQC track, rather than as a separate concern. And the study design — particularly the open ETSI 014 emulator and the published agility rubric — gives the quantum-information research community a reproducible substrate for follow-on measurement, critique, and refinement, on a single commodity host.

The work has not been published or submitted elsewhere. All authors have read and approved this submission. We declare no competing interests; the work was self-funded by E2E Solutions and Hacettepe University with no external grants. The data and code-availability statement in the manuscript points to a fully open repository covering the schema, the implementation, the replay corpus, the ruleset, the hand-graded ground-truth corpus, and the Tranco snapshot identifier used for the Study 1 sample frame.

We thank the editorial team and the reviewers for their consideration.

Sincerely,

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